Calculus I Homework 4

Department of Mathematics

Instructions

Show all work and simplify final answers. You may use a calculator for arithmetic only.

Problems

1. Compute the limit, or state that it does not exist.

$$\lim_{x \to 2} \frac{x^2 - 4}{x - 2}$$

$$\lim_{x \to 2} \frac{x^2 - 4}{x - 2} = \lim_{x \to 2} \frac{(x-2)(x+2)}{x - 2} = \lim_{x \to 2} (x+2) = \boxed{4}$$

2. Find the derivative of the function.

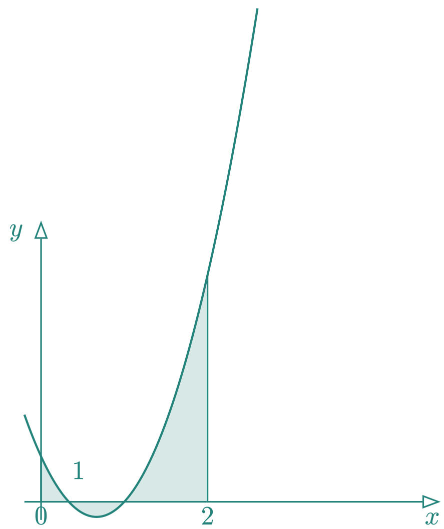
$$f(x) = x^3 \ln(x) - \frac{4}{x^2}$$

$$f(x) = x^3 \ln(x) - \frac{4}{x^2} = x^3 \ln(x) - 4x^{-2}$$

$$f'(x) = \left(x^3 \ln(x)\right)' - 4\left(x^{-2}\right)' = 3x^2 \ln(x) + x^3 \cdot \frac{1}{x} + 8x^{-3} = \boxed{3x^2 \ln(x) + x^2 + \frac{8}{x^3}}$$

3. Compute the definite integral and sketch $y = 3x^2 - 4x + 1$ with the area for the integral shaded.

$$\int_0^2 (3x^2 - 4x + 1)\,dx$$



$$\int_0^2 (3x^2 - 4x + 1)\,\mathrm{d}x = \left[x^3 - 2x^2 + x\right]_0^2 = (8 - 8 + 2) - 0 = \boxed{2}$$

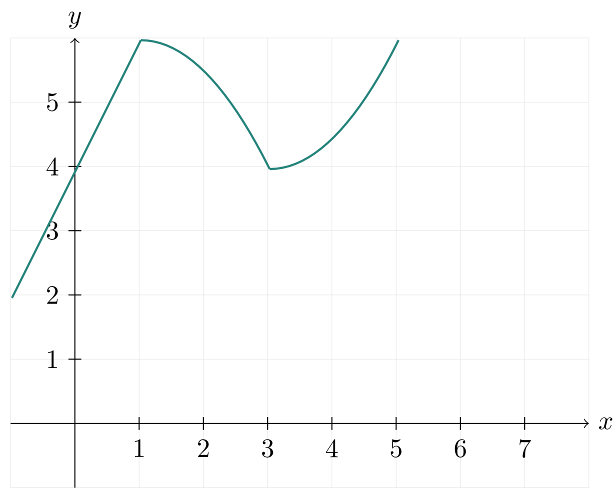
4. Let $f(t)$ be defined by

$$f(t) = \begin{cases} 2 & 0 \le t < 2, \\ 2 - t & 2 \le t < 4, \\ t - 4 & 4 \le t \le 6. \end{cases}$$

Define

$$F(x) = \int_0^x f(t)\, dt.$$

On the axes below, sketch the graph of $F(x)$ for $0 \le x \le 6$.



5. Use the Fundamental Theorem of Calculus to find the derivative.

$$G(x) = \int_1^x \sqrt{1 + t^4}\, dt$$

$$G'(x) = \sqrt{1 + x^4} \cdot \frac{\mathrm{d}}{\mathrm{d}x}(x) = \boxed{\sqrt{1 + x^4}}$$